

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

In the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. The point $(6, c)$, where c is a constant, lies on this circle. What is the value of c ?

Correct Answer: 5

Rationale

The correct answer is **5**. It's given that in the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. It's also given that the point $(6, c)$, where c is a constant, lies on this circle. It follows that the ordered pair $(6, c)$ makes the equation $(x - 3)^2 + (y - 5)^2 = 9$ true. Substituting **6** for x and c for y in this equation yields $(6 - 3)^2 + (c - 5)^2 = 9$, or $9 + (c - 5)^2 = 9$. Subtracting **9** from each side of this equation yields $(c - 5)^2 = 0$. It follows that the value of c is **5**.